

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	05-07-2026
Team ID	SWTID-2026-8108
Project Name	Heart Disease Analysis
Maximum Marks	

Technical Architecture:

For your specific project, the system flow moves through three core architectural tiers:

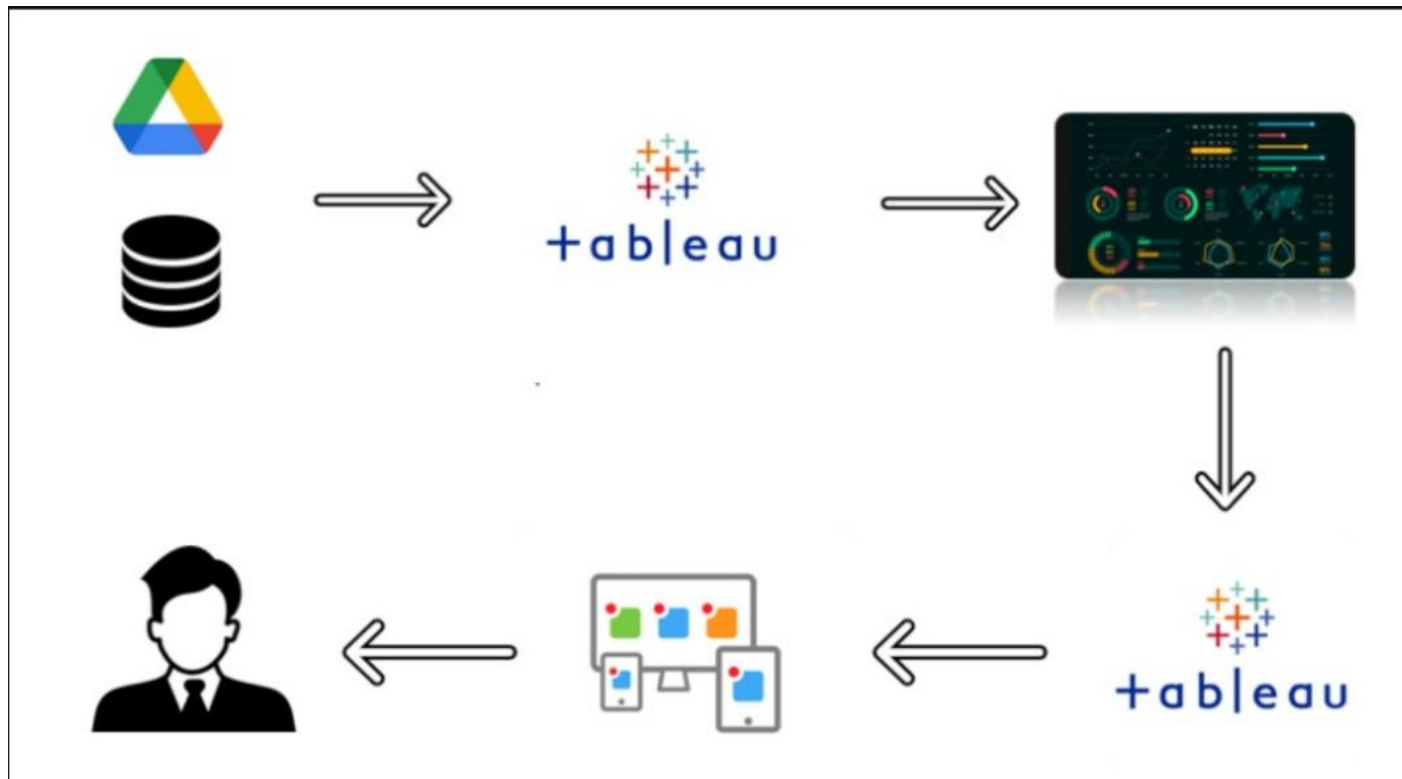


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interactive visualization interface used by Dr. Sharma, Ramesh, and Anita to cross-filter clinical metrics.	Tableau Desktop / Tableau Storyboard Canvas
2.	Application Logic-1	Logic used to calculate multi-variable aggregations, group distributions, and risk trends across data variables.	Tableau Calculated Fields & Aggregation Functions
3.	Application Logic-2	Structured queries executed to isolate specific comorbidity groups and sort patient lifestyle conditions.	Native MySQL SQL Query Scripts
4.	Database	Relational database schema containing the complete cleaned heart disease risk factors dataset.	MySQL 8.0 Relational Database Engine
5.	File Storage	Storage of the primary dataset file and project dashboard files locally.	Local Filesystem (.CSV Data File & .TWBX Workbook File)
6.	Infrastructure (Server / Cloud)	Environment used for hosting the database engine and rendering individual dashboard panes.	Local Server Configuration: Localhost environment powered by MySQL InnoDB Database Server.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source relational infrastructure used to maintain structural database operations across columns.	MySQL Community Server Open-Source Database Engine
2.	Security Implementations	Local data restrictions implemented to safeguard health records from unverified extraction attempts.	MySQL User Authentication Controls & Secure Local Database Password Encryptions
3.	Scalable Architecture	Multi-tier application architectural design separating raw data storage from visual interface layers.	2-Tier Client-Server Architecture (MySQL Database Tier linked directly to Tableau Presentation Tier)
4.	Availability	Ensuring persistent local dashboard access throughout live deployment runs and user sessions.	MySQL
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Tableau Hyper Data Extraction Layer & MySQL Indexing optimizations for rapid query computation